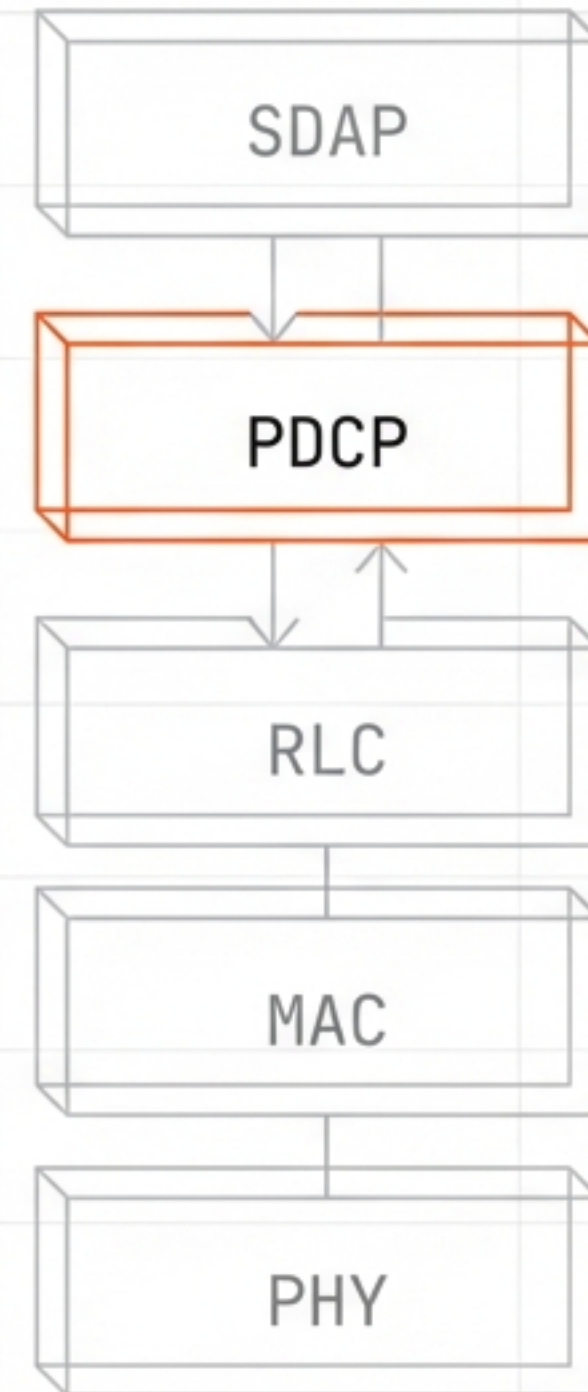


5G NR PDCP: Integrity Protection and Verification

Technical Analysis of 3GPP
TS 38.323, Section 5.9

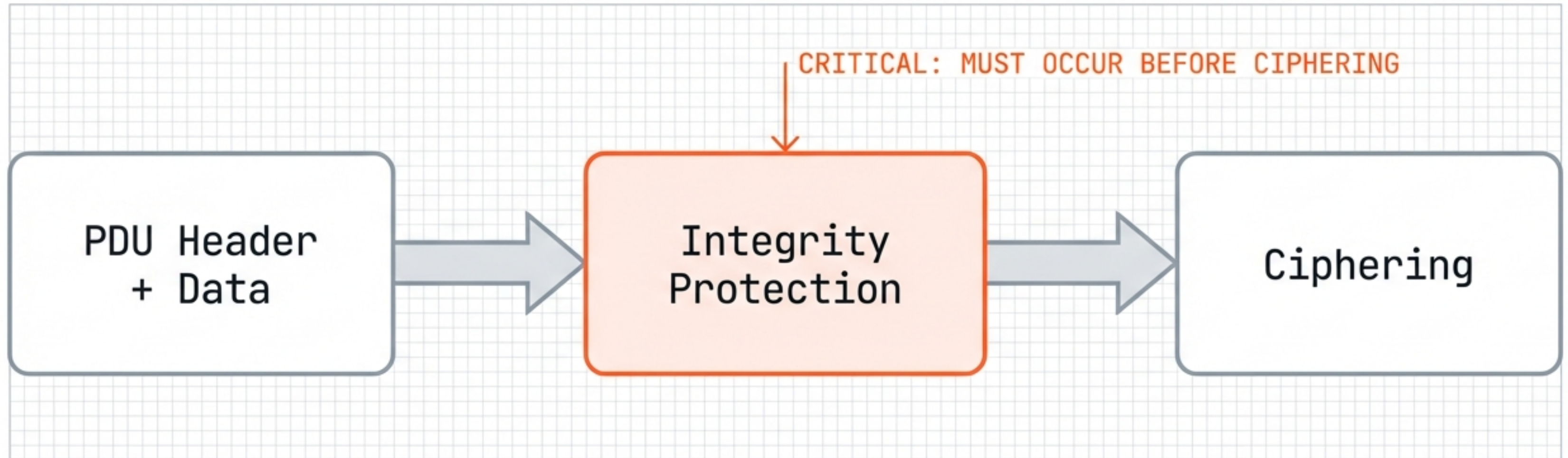
Document Version 19.0.0 Context

Prepared for Technical Engineering Review



The Golden Rule

Integrity Protection sequence is rigid and non-negotiable.



In the 5G protocol stack, the PDCP layer acts as the security gatekeeper. Integrity protection ensures data has not been tampered with during transit. This function is mathematically applied to the header and payload prior to encryption.

Applicability Matrix

Control Plane (SRBs)



MANDATORY

Integrity protection is always applied to PDCP Data PDUs.

Signalling Radio Bearers:
SRB1, SRB2, SRB3 (Sidelink).

User Plane (DRBs)



CONFIGURABLE

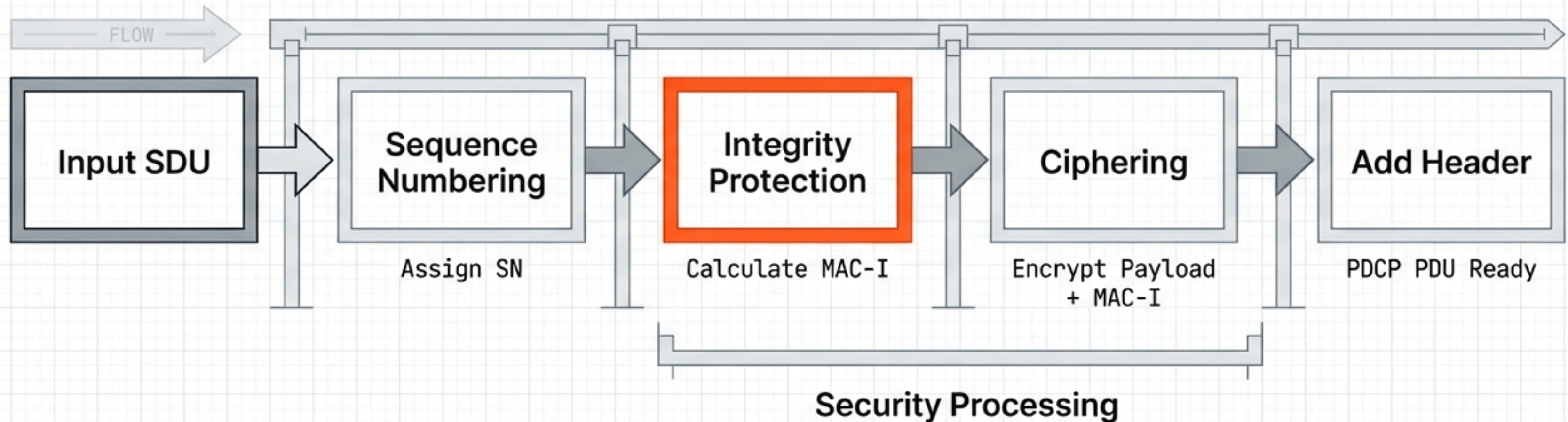
Integrity protection is applied only if explicitly configured by RRC.

Data Radio Bearers:
Configured per session.

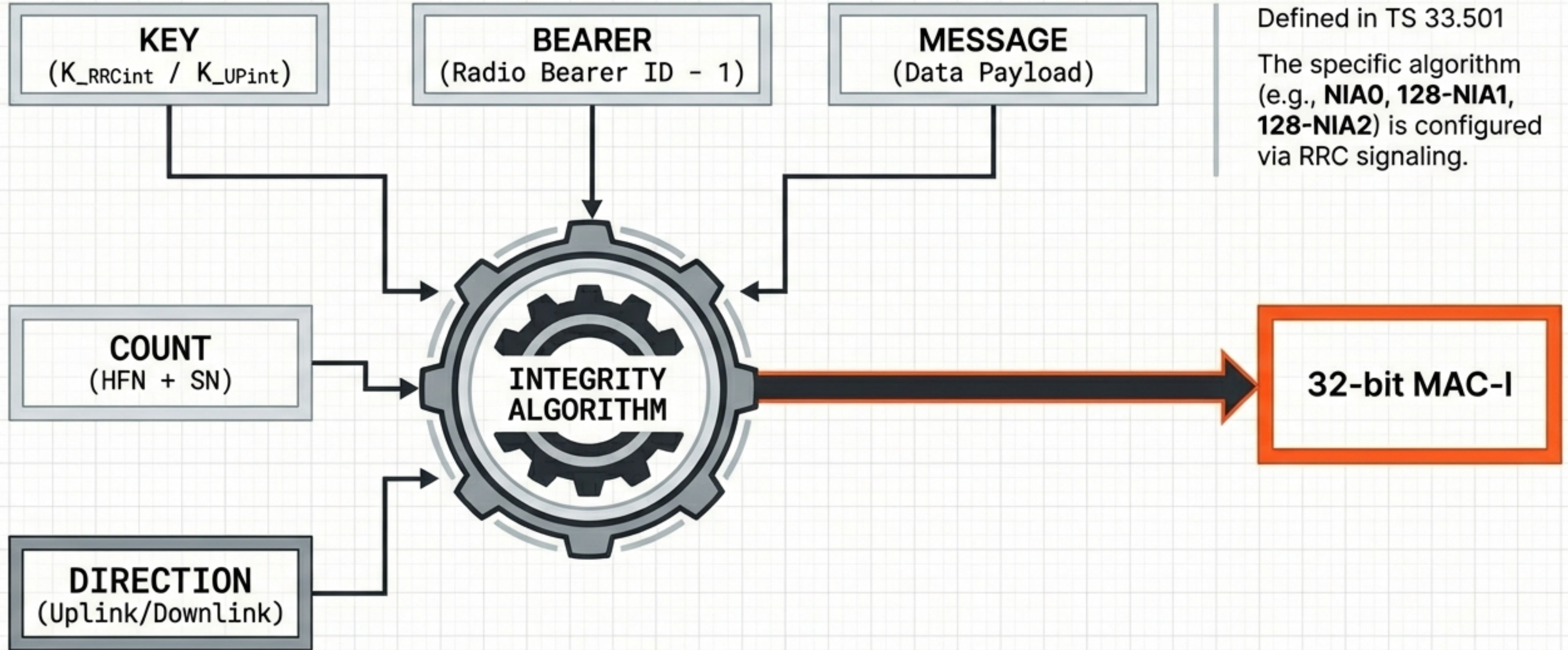
EXCLUSIONS: PDCP Control PDUs | Multicast/Broadcast Bearers (MRBs) | Sidelink SRB4

Transmitter Pipeline: Sequence of Operations

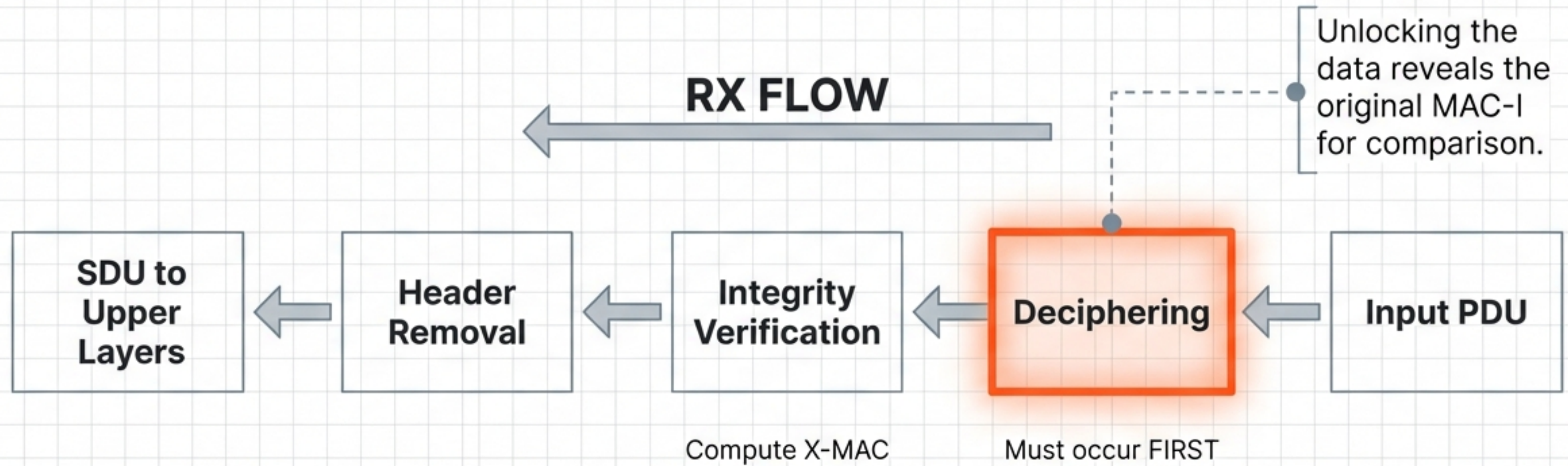
The chronological transformation of a Service Data Unit (SDU) into a Protocol Data Unit (PDU). Note that the MAC-I is generated before the packet is encrypted.



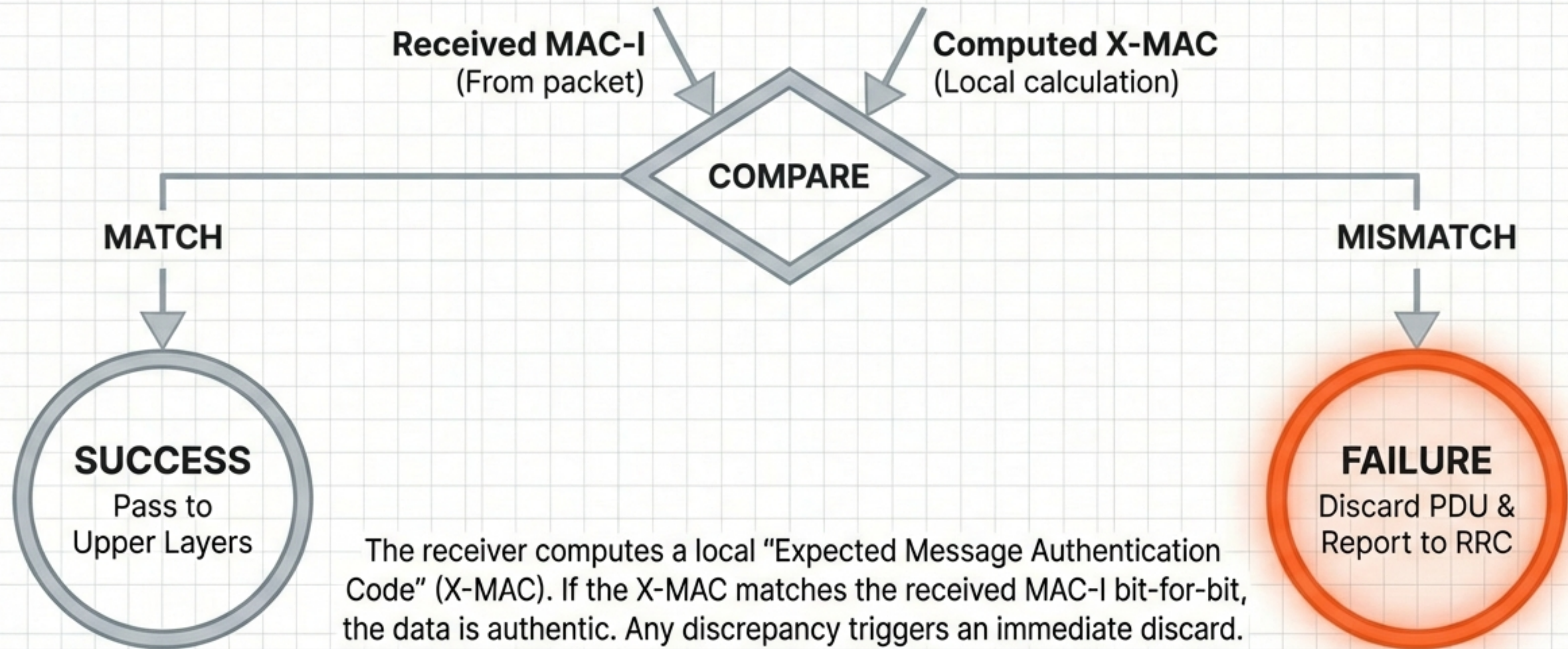
Deep Dive: MAC-I Calculation Algorithm



Receiver Pipeline: Reverse Engineering



Verification Logic: The X-MAC Decision



Special Scenarios



NR Sidelink (V2X)

Applies to Sidelink SRBs and Unicast DRBs.

If protection is not required, header field 'KNRP-sess ID' is set to 0.



DAPS (Dual Active Protocol Stack)

During handover, connection is maintained with Source and Target simultaneously.

PDPC entity manages two security contexts. Keys/Algorithms applied based on target cell.

System Summary: Integrity Protection

SCOPE

Protects PDU Header + Data Payload.

ORDER

Tx: Integrity BEFORE Ciphering.
Rx: Deciphering BEFORE Integrity.

TOKEN

32-bit MAC-I (Tx) vs. 32-bit X-MAC (Rx).

APPLICABILITY

Mandatory: SRB 1, 2, 3.
Configurable: DRBs.

INPUTS

Key, COUNT (HFN+SN), Direction, Bearer ID, Message.

FAILURE PROTOCOL

Immediate PDU discard.
Indication to upper layer.

References & Specifications

Primary Source:

3GPP TS 38.323 V19.0.0

Packet Data Convergence Protocol (PDCP) Specification

Secondary Source:

3GPP TS 33.501

Security Architecture and Procedures for 5G System

End of Document.